Complex	$[Fe(CN)_6]^{3-}$
Outer orbitals of metal atom/ion in presence of ligands	$3d^5$: ↑↓ ↑↓ ↑ ; $4s^0$: ; $4p^0$:
Hybridisation	Coordination number - 6 Hybridsation - d^2sp^3
Hybridised orbitals of the metal ion in the complex	$3d^5$: ↑↓ ↑↓ ↑ ; d^2sp^3 Hybridised orbitals : ↑↓ ↑↓ ↑↓ ↑↓ ↑↓ ↑↓
Geometry	Octahedral In this complex inner d orbitals are involved in hybridisaion and hence the complex is called inner orbital complex
Magnetic property	No. of unparied electrons = 1; Hence paramagnetic
Magnetic moment (Using spin only formula)	$\mu_s = \sqrt{n(n+2)} = \sqrt{1(1+2)} = 1.732$ BM

Illustration 4

Complex	$[CoF_6]^{3-}$
Central metal atom and its outer electronic configuration	Co^{3+}: $3d^6$, $4s^0$
Outer orbitals of metal atom/ion	$3d^6$: ↑↓ ↑ ↑ ↑ ↑ ; $4s^0$: ; 4^p :
Nature of ligand	F^- Weak field ligand and hence no pairing of 3d electrons in the metal
Outer orbitals of metal atom/ion in presence of ligand	$3d^6$: ↑↓ ↑ ↑ ↑ ↑ ; $4s^0$: ; $4p^0$: ; $4d^0$:
Hybridisation	Coordination number - 6 Hybridsation - sp^3d^2

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